A BALANCE-MODE STRUCTURE THEOREM FOR THE UNSIGNED KERNEL OF THE SPHENIC WINDOW COMPLEX

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Abstract. Let C be the sphenic window complex at level N: the 2-complex whose triangles are the squarefree three-prime integers pqr in $W = (N/2, N]$, with edges the co-occurring prime pairs and vertices the primes. Let ∂^u be its *unsigned* boundary, the all-positive incidence map, which presents C as an oriented hypergraph with every adjacency negative in the sense of Rusnak. We describe the rational kernel of ∂^u as a span of explicit *balance modes* fibered over vertex-pairs, and certify it at an explicit finite perimeter. For the five bundled complexes $N \in \{50000, 100000, 200000, 300000, 500000\}$, certified over $\mathbb{Q}$ and an eight-prime battery at all five N (and over every prime characteristic outside $\{2, 3\}$ on the two-stage integer-certified subset $N \in \{10^5, 2 \times 10^5\}$),

$$\ker \partial^u = \operatorname{span}\left\{ m_{a,b,z} = \textstyle\sum_g z_g([a \cup g] - [b \cup g]) : a < b \text{ prime},\ z \in Z_{\text{bal}}(L_{a,b}) \right\},$$

where $L_{a,b}$ is the common link graph of the pair and Z_{bal} its balance kernel (the Zaslavsky even-circle cycle space). Consequently, at that perimeter, $e := \dim \ker \partial^u$ is a fibered balance count, $\text{ru} = F - e$, the frustration term is $\delta = \text{elem} - e$, and the canonical density $\kappa = (b_1 - \delta)/V$ is determined by link-graph balance and connectivity combinatorics. The characteristics 2 and 3 are exceptional and exactly understood: over $\mathbb{F}_2$ the kernel is elem, over $\mathbb{F}_3$ it is $e + 1$. The generators are the classified balanced and balanceable circuits of the oriented-hypergraph program of Rusnak [3] and of Rusnak–Li–Xu–Yan–Zhu [4]; the apparently unstated *spanning* statement addresses their declared open problem. We prove the reduction unconditionally, reduce the remaining content to a single finite leakage-gauge condition (LG), and certify (LG) at the perimeter above. The all-N form is [open], with the band localization proved and a richness program stated.

1. The unsigned kernel, and the statement at the certified perimeter

1.1. The object. The complex C is as in the companion Paper 3: triangles are the sphenics $pqr \in (N/2, N]$, edges the co-occurring prime pairs, vertices the primes. The signed boundary ∂_2 has rank $F - \text{elem}$ by the slab identity. This paper is about the *unsigned* boundary

$$\partial^u[p, q, r] = [q, r] + [p, r] + [p, q] \qquad (\text{all coefficients } +1),$$

the incidence map of C as an oriented hypergraph with all-positive incidence, equivalently all adjacencies negative in Rusnak's dictionary. Over a field of characteristic $\neq 2$ the two boundaries differ, and their nullities differ by the frustration term

$$\delta := \text{ru} - \operatorname{rank} \partial_2 = (F - e) - (F - \text{elem}) = \text{elem} - e, \qquad e := \dim \ker \partial^u, \quad \text{ru} := \operatorname{rank} \partial^u.$$

The term δ is the one quantity in the whole coprime-graph arc that has resisted a rank-free closed form; the point of the structure theorem is to express e, hence δ, as a balance count.

1.2. The fiber objects. For a vertex v, the *link* L_v is the graph on primes with an edge $\{x, y\}$ whenever $\{v, x, y\}$ is a triangle of C. For a prime pair $\{a, b\}$, the *common link* $L_{a,b}$ is the graph whose edges $g = \{x, y\}$ satisfy that both $\{a, x, y\}$ and $\{b, x, y\}$ are triangles. Signing every adjacency negative (equivalently, the all-positive incidence), the *balance kernel* $Z_{\text{bal}}(G)$ is the Zaslavsky even-circle cycle space [5]: the space spanned by the even circles and the odd-circle barbells of G,

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the cycle space of the signed graphic matroid of the all-negative signing. For each pair and each $z \in Z_{\mathrm{bal}}(L_{a,b})$, the *balance mode* is $m_{a,b,z} = \sum_g z_g([a \cup g] - [b \cup g])$, with g ranging over the edges of $L_{a,b}$.

1.3. The theorem, scoped.

Structure theorem (unsigned kernel; certified perimeter). [BANKED AS C-0746 AT THE CERTIFIED PERIMETER] *For each of the five bundled complexes $N \in \{50000, 100000, 200000, 300000, 500000\}$, certified over $\mathbb{Q}$ and an eight-prime battery at all five N, and over every prime characteristic $p \notin \{2, 3\}$ on the two-stage integer-certified subset $N \in \{10^5, 2 \times 10^5\}$,*

$$\ker \partial^u = \mathrm{span}\{\, m_{a,b,z} : a < b \text{ prime},\ z \in Z_{\mathrm{bal}}(L_{a,b})\,\}.$$

We state the scope plainly. The inclusion $\supseteq$ (the modes are kernel vectors) is an unconditional lemma for all N (§2); the reduction of $\subseteq$ to a finite condition (LG) is unconditional for all N (§3). What is verified rather than proved in general is (LG) itself, verified at the five N by an exact certificate stack (§4). The characteristic scope matches the certificate inventory exactly: the leakage-gauge two-rank test over an eight-prime battery covers all five N and pins the kernel over $\mathbb{Q}$, while the all-characteristic closure (no bad-prime window) is certified by the two-stage integer certificates only on $N \in \{10^5, 2 \times 10^5\}$; we do not claim all characteristics at the three larger N, where those integer certificates are not yet in the repository. The all-N statement is open (§7), and every downstream consequence inherits the certified-perimeter scope.

2. The proved core: modes are cycles, and the cone split

Lemma A (modes are cycles). [PROVED, ALL N] *For any prime pair $\{a, b\}$ and any $z \in Z_{\mathrm{bal}}(L_{a,b})$ (zero unsigned vertex sums on $L_{a,b}$), $\partial^u m_{a,b,z} = 0$.*

Proof. Examine each facet coefficient of $\partial^u m$. A facet $\{x, y\}$ interior to an edge g receives z_g from $[a \cup g]$ and z_g from $[b \cup g]$ with the mode's sign difference, so the coefficients cancel. A facet $\{a, u\}$ receives $\sum_{g \ni u} z_g$, the unsigned vertex sum of z at u, which is zero by balance; likewise $\{b, u\}$. The facet $\{a, b\}$ never occurs. Hence $\partial^u m = 0$. □

Filter the kernel by least vertex. For a vertex v let $C_{\geq v}$ be the subcomplex on vertices $\geq v$, and let $w \in \ker \partial^u|_{C_{\geq v}}$ have least vertex v. Split $\mathrm{supp}(w) = A \sqcup B$, A the cells containing v, and write w_A for the tail chain of the A-cells.

Lemma B (cone split at the least vertex). [PROVED, ALL N; CHARACTERISTIC $\neq 2$] (i) *$w_A \in Z_{\mathrm{bal}}(L_v^{\geq v})$ (the v-facet block);* (ii) *$w_A = -\partial^u(u)$ exactly, where u is the B-part of w (the v-free block, both sides supported on L_v-edges);* (iii) *if $B = \varnothing$ then $w = 0$.*

Writing W_v for the space of A-parts of kernel vectors of $C_{\geq v}$ with least vertex v, we obtain $W_v = Z_{\mathrm{bal}}(L_v^{\geq v}) \cap \partial^u(\text{chains of } C_{>v})$. The second condition is not decorative: the naive version without it is false, with a measured deficit of 1276 at $N = 10^5$; any proof ignoring the boundary witness is refuted by that counterexample.

Lemma C (graded reduction). [PROVED, ALL N] *The A-part map identifies the graded quotient $K(C_{\geq v})/K(C_{>v})$ with W_v, and the modes with $a = v$ have A-parts exactly $\sum_{b>v} Z_{\mathrm{bal}}(L_{v,b}^{\geq v}) \subseteq W_v$. Hence, by downward induction on the finitely many least-vertex levels (primes up to roughly $N^{1/3}$), the structure theorem for $C_{\geq v}$ follows from the theorem for $C_{>v}$ together with the following.*

Realizable Decomposition Lemma (the crux). *$W_v \subseteq \sum_{b>v} Z_{\mathrm{bal}}(L_{v,b}^{\geq v})$ at every level: every balanced, link-supported boundary of the upper subcomplex splits into pair-fiber balance cycles.*

Lemmas A, B and C are unconditional; the whole remaining content is the crux, which §3 reduces to a finite gauge condition.

3. The peeling architecture, with corrected bookkeeping

We prove the crux under a finite hypothesis (LG). The essential correction over earlier attempts is the bookkeeping: one does *not* hold the witness boundary fixed while removing a layer; each peeling step subtracts a certified element q from the current boundary, so the boundary itself changes as $h \mapsto h - q$.

3.1. **Notation.** Order vertices by prime value. For $a > v$ let X_a be the cells with all vertices $\geq a$, and a^+ the next vertex. For $b \geq a$ let $L^a_{v,b}$ be the common link on vertices $\geq a$ with b removed, and $T^a_v = \sum_{b \geq a} Z_{\mathrm{bal}}(L^a_{v,b})$, so $T^{a_0}_v = T$, the target pair-fiber span (a_0 the first vertex above v). Let L_a be the link of a inside X_a and $T_a = \sum_{c>a} Z_{\mathrm{bal}}(L_{a,c})$. For $u \in C_2(X_a)$, its a-tail $\ell_a(u)$ keeps the tail $[x, y]$ of each cell $[a, x, y]$ with least vertex a and sends all other cells to 0.

3.2. **Mode halves and the leakage tail.** For $z \in Z_{\mathrm{bal}}(L^a_{v,b})$, the C'-half of the full v-mode is $R_{v,b}(z) = -\sum_g z_g[b \cup g]$, and balance gives $\partial^u R_{v,b}(z) = -z$ on the link edges. Summing over b, for $\mathbf{z} = (z_b)$ one gets $R(\mathbf{z})$ with $\partial^u R(\mathbf{z}) = -q(\mathbf{z})$, where $q(\mathbf{z}) = \sum_b z_b \in T^a_v$. The *leakage-tail map* is $\beta_{v,a}(\mathbf{z}) = \ell_a(R(\mathbf{z}))$, an explicit linear map into $C_1(L_a)$: the tail edge $\{x, y\}$ receives $-(z_a)_{xy} - (z_x)_{ay} - (z_y)_{ax}$, absent terms read as 0.

3.3. **Straddle localization by the window (P1).** Let $u \in C_2(X_a)$ with $h = \partial^u u \in Z_{\mathrm{bal}}(L^a_v)$ and $t = \ell_a(u)$. The imbalance of t at x equals h_{ax} when $\{a, x\}$ is a link edge and 0 otherwise, so it is supported on the straddle set $S_{v,a} = \{x > a : \{a, x\} \in L_v\}$ and sums to zero there. The window supplies (P1): if a cell $\{a, x, y\}$ shares the edge $\{a, x\}$ with a cell $\{v, a, x\}$, their third vertices y, v differ by a factor below 2, so $y < 2v$. In particular, once $a \geq 2v$ no straddling contribution occurs and the a-tail is automatically balanced, confining the difficulty to the factor-2 band $a \in (v, 2v)$.

3.4. **The finite leakage-gauge hypothesis.** Let $\mathcal{U}_{v,a} = \{\ell_a(u) : u \in C_2(X_a),\ \partial^u u$ supported on and balanced in

Hypothesis (LG)$_{v,a}$. For every $t \in \mathcal{U}_{v,a}$ there is $\mathbf{z}$ with $t + \beta_{v,a}(\mathbf{z}) \in T_a$; equivalently, in $C_1(L_a)/T_a$, $\overline{\mathcal{U}}_{v,a} \subseteq \operatorname{im} \overline{\beta}_{v,a}$. A stronger link-only sufficient form is $C_1(L_a) = \operatorname{im} \beta_{v,a} + T_a$.

3.5. **One-step elimination and the induction.**

One-step elimination lemma. [proved under (LG)$_{v,a}$] *Given $u \in C_2(X_a)$ with $h = \partial^u u \in Z_{\mathrm{bal}}(L^a_v)$, there exist $q \in T^a_v$ and $u^+ \in C_2(X_{a^+})$ with $\partial^u u^+ = h - q$ and $h - q \in Z_{\mathrm{bal}}(L^{a^+}_v)$.*

Proof. Let $t = \ell_a(u)$. By (LG)$_{v,a}$ pick $\mathbf{z}$ with $p := t + \beta_{v,a}(\mathbf{z}) \in T_a$, and set $R = R(\mathbf{z})$, $q = q(\mathbf{z}) \in T^a_v$, so $\partial^u R = -q$ and $\ell_a(R) = \beta_{v,a}(\mathbf{z})$. Since internal kernel tails are exactly T_a (Lemma C inside X_a), choose an internal cycle N with $\ell_a(N) = p$. Put $u^+ = u + R - N$. Then $\partial^u u^+ = h - q$, and $\ell_a(u^+) = t + \beta_{v,a}(\mathbf{z}) - p = 0$, so u^+ has no cell with least vertex a and lies in $C_2(X_{a^+})$. Finally $h - q$ is balanced and, being $\partial^u u^+$ with u^+ free of a, has no edge incident to a, so $h - q \in Z_{\mathrm{bal}}(L^{a^+}_v)$. $\square$

Reverse induction on a proves: if $u \in C_2(X_a)$ and $\partial^u u = h \in Z_{\mathrm{bal}}(L^a_v)$, then $h \in T^a_v$. The base after the largest vertex is trivial; the step is the one-step lemma plus the induction hypothesis on $h - q$. Taking $a = a_0$ gives $W_v \subseteq T$, the crux, hence the structure theorem under the (LG) hypotheses. Unwinding exhibits the corrected bookkeeping: chains $u_0 = u, u_1, \ldots, u_r = 0$ and certified $q_i \in T$ with $\partial^u u_i = h_i$, $h_{i+1} = h_i - q_i$, so $h = q_0 + \cdots + q_{r-1} \in T$. The witness boundary is not held fixed; each peel subtracts a certified pair-fiber element.

4. (LG) as a finite condition, and the certificate system

4.1. **The finite rank form.** For fixed a, (LG)$_{v,a}$ is a single rank equality. Let U_a span the occurring tails $\mathcal{U}_{v,a}$ (a kernel of an explicit two-block boundary matrix), G_a span $\operatorname{im} \beta_{v,a}$, and H_a span T_a. Then

$$(\mathrm{LG})_{v,a} \text{ holds} \iff \operatorname{rank}[\,G_a\ H_a\ U_a\,] = \operatorname{rank}[\,G_a\ H_a\,],$$

a machine-checkable equality of ranks of explicit integer matrices, one per level pair. The band localization confines the nontrivial checks to $a \in (v, 2v)$.

4.2. The certificate stack. The structure theorem is certified at the five bundled complexes by the following, all reproducible from the repository root (bundles under `certificates/` are gzipped JSON with SHA-256 manifests).

- *Leakage-gauge rank certificates.* The two-rank test passes at 616 level pairs across the five N, integer-exact. [CERTIFIED]
- *Eight-prime battery at emission.* Every rank confirmed over eight prime characteristics as certificates are emitted. [CERTIFIED]
- *Modular sweep.* A 32,868-check sweep over characteristics up to 997 finds no bad prime below 1000. [VERIFIED]
- *Two-stage integer certificates.* For $N \in \{10^5, 2\times 10^5\}$ (198 pairs), a unit-pivot unimodular reduction followed by an exact Smith normal form of the residual block produces integer invariant factors with none divisible by any prime outside $\{2, 3\}$, closing the field story on that subset without a bad-prime window above 1000; at the three larger N the certificates cover $\mathbb{Q}$ and the eight-prime battery, not all characteristics. [CERTIFIED ON $N \in \{10^5, 2\times 10^5\}$]
- *Independent enumeration replay.* A separate program rebuilds every cell inventory from N alone and reproduces the fiber and generator data. [VERIFIED CLEAN]
- *Independent checker replay.* A second implementation, using spanning-tree balance bases and a different elimination order, re-verifies every rank over the battery. [VERIFIED CLEAN]

The formal adjudication banked the $\mathbb{Q}$ plus eight-prime perimeter at $N \in \{10^5, 2\times 10^5\}$ (198 pairs, 1584 rank-pair checks); the five-N, 616-certificate extension is on the same epistemic footing, a theorem modulo a hashed finite certificate.

4.3. The bug that failed loudly (reproducibility). The independent checker earned its keep. Its first version had a sign error in the barbell balance-closure, and it failed loudly: the checker disagreed with the emitter on 22 of 78 fibers, at all eight primes at once. The fault was in the checker, not the certificates, and a loud, characteristic-independent disagreement is exactly what an independent replay should produce when something is wrong. The checker was rewritten with a brute-forced sign closure plus a per-component dimension guard (each balance component contributes $E - V + [\text{bipartite}]$), after which emitter, checker and enumerator agree on every pair. A certificate stack whose independent replays never disagree has not really been tested; the loud disagreement is the evidence that the replay is genuinely independent, and the adjudication credited it as such.

5. Characteristic stratification

The kernel dimension of ∂^u is not field-independent, and the two exceptional characteristics are exactly the ones the oriented-hypergraph taxonomy predicts.

- *Characteristic* 2 [PROVED; FROM C-0744]: over $\mathbb{F}_2$ the unsigned and signed boundaries coincide, so $\ker \partial^u = \ker \partial_2$ has dimension elem (the fitting-cell space, not the balance-mode space).
- *Characteristic* 3 [VERIFIED; CERTIFIED ON THE PERIMETER]: over $\mathbb{F}_3$ the kernel dimension is exactly $e + 1$, one extra cross-theta class. This is the all-entrant $\mathrm{GF}(k)$ phenomenon: an all-entrant minimal k-cross-theta is minimally dependent only over $\mathrm{GF}(k)$, so at $k = 3$ exactly one such dependency becomes rational over $\mathbb{F}_3$ and disappears elsewhere. The measured $e + 1$ is a confirmed prediction of Rusnak's characteristic theory.

- *Every other characteristic* [CERTIFIED ON THE PERIMETER]: over $\mathbb{Q}$ and over every prime outside $\{2, 3\}$ the kernel dimension is e (two-stage subset; battery-scoped elsewhere), by the certificate stack of §4.

On the certified set the picture is a clean stratification: $\mathbb{F}_2$ gives elem, $\mathbb{F}_3$ gives $e + 1$, and the generic e holds everywhere else, with $\delta = \text{elem} - e$ the characteristic-2 jump.

6. Positioning in the oriented-hypergraph program

The generators are classified circuits; only the spanning statement is ours to claim, and even that is scoped. Under the standard dictionary (all-positive incidence corresponds to all-negative adjacency, so a positive circle is an even circle and Zaslavsky balance is evenness), the two mode families are known objects. The parity boxes are *balanced circuits* in the sense of Rusnak, *Oriented Hypergraphs: Introduction and Balance* [3] (the arXiv preprint `1210.0943` carries the title *Oriented Hypergraphs I*); their classification of balanced circuits is the relevant result (Theorem 7.2.7 in the arXiv numbering, to be confirmed against the published version before quoting the number). The barbell modes are *balanceable circuits* in the sense of Rusnak, Li, Xu, Yan and Zhu, *Oriented Hypergraphs: Balanceability* [4]. We cite Lucas J. Rusnak as sole author of the 2013 paper.

What is apparently unstated is the spanning statement. No nullity or spanning theorem exists in the oriented-hypergraph program: the unbalanceable circuit family is their declared open problem, and no global kernel-dimension formula for hypergraph incidence matrices is known at all. Our theorem, restricted to the sphenic/window family and at the certified perimeter, asserts that the two classified families already span the rational kernel, fibered over vertex-pairs by link balance; equivalently, the unbalanceable circuits are rationally redundant on this family. There is independent structural support: the all-entrant minimal k-cross-thetas are minimally dependent only over $\mathrm{GF}(k)$, so the simplest unbalanceable circuits contribute no rational kernel vectors, exactly the mechanism visible as the $\mathbb{F}_3$ stratum of §5. We frame the contribution as an apparently unstated spanning equality over their taxonomy, not as a new circuit theory. The fiber object is the Zaslavsky even-circle matroid, suggesting a matroid-union reading of e (a route, not a result). Adjacent but distinct is the Nagel–Reiner complex-of-boxes machinery [7], a different functor on the same skew objects; the one named residual specialist check is Duval's relative Laplacian spectral recursion [6].

7. The all-N problem, and what is proved toward it

The all-N structure theorem is [OPEN]. The obstruction is precisely a uniform (LG). Two things are proved toward it. *Band localization* [PROVED]: by (P1), all straddle imbalance lives in the factor-2 band, so for a level $a \geq 2v$ every occurring a-tail is automatically balanced and the nontrivial gauge condition is confined to partners $a \in (v, 2v)$; above $2v$ the peeling is straddle-free and unconditional. *The richness program* [OPEN, SHARPLY STATED]: what remains is to show the leakage map is surjective onto the occurring band tails uniformly in N, plausibly under an explicit finite richness hypothesis on the band fibers (enough independent even circles and barbells in each $L_{v,b}$ with $v < b < 2v$ to realize every occurring straddle correction). This is a concrete, machine-testable family of conditions; proving it uniformly, or replacing it by a band-local witness-modification argument, would upgrade the theorem from the certified perimeter to all N.

8. Consequences at the certified perimeter

At the certified perimeter the whole coprime-graph arc becomes counting. The unsigned nullity e is the dimension of a sum of pair-fiber balance spaces, computed by inclusion-exclusion over overlapping pairs; each component's balance dimension is $E - V + [\text{bipartite}]$, with the non-bipartite fraction growing with scale (measured by the bipartite census at three of the certified N: $0.73, 0.81, 0.88$), confining corrections to the completer fringe. The frustration term $\delta = \text{elem} - e$ becomes the characteristic-2 jump of a certified quantity, and the canonical density $\kappa = (b_1 - \delta)/V$

is reduced, at these N, to link-graph balance and connectivity combinatorics. Combined with the exact signed profile of the companion Paper 3, this removes the last uncertified rank from the sphenic ($\omega = 3$) layer of the author's -1-multiplicity decomposition [1, 2] at the perimeter; the present paper is the fourth in that series. The transcendence question for κ at infinity remains tied to the all-N richness program.

Appendix A. Machine certificates (reproducible)

All scripts run from the repository root (concept DOI `10.5281/zenodo.21135221`). `mode_completeness.py` (and a 5×10^5 variant): global completeness, rank of the mode family equals e at $N \in \{5 \times 10^4, 10^5, 2 \times 10^5, 5 \times 10^5\}$, values $482, 1270, 3082, 9402$. `graded_completeness.py`: the graded crux verified level by level. `atomic_space_test.py`: the atomic realizable decomposition at space level. `lg_rank_test.py`, `twostage_certify.py`: the (LG) two-rank and the two-stage integer certificates. `lg_replay_check.py`, `enum_replay.py`: the independent checker and enumeration replays. `gauge_anatomy2.py`, `slab_gf2_sweep.py`: the $\mathbb{F}_3$ and $\mathbb{F}_2$ characteristic strata.

Acknowledgement / disclosure

The balance-mode reformulation of the unsigned kernel and the reduction of completeness to the finite leakage-gauge condition (LG) are the author's; the one-step elimination architecture of §3 originated in an AI-assisted solver round and was independently cold-refereed. The manuscript was prepared with assistance from a large language model (Claude, by Anthropic) for organisation, the classical background and prior-art positioning, the balance formalism, and extensive symbolic and numerical verification; all statements, proofs, and their interpretation are the author's and were checked independently, with banking-grade claims filed through the laboratory's adversarial adjudication protocol before being marked proved. Results carry status labels: [PROVED], [BANKED], [VERIFIED], [CERTIFIED], [OPEN].

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